

Project 2 — SQL Analysis: Customer Profitability (Logistics)

Case Study (PostgreSQL)

Business Problem

- Logistics teams often see high-volume customers as top performers, but volume does not always mean profit. This project identifies profitable vs loss-making customers and explains the drivers behind profitability (cost structure, shipment weight, and service level).

Objective

- Build a repeatable SQL workflow that calculates shipment-level profit, aggregates to customer scorecards, ranks customers, and evaluates the impact of late deliveries over time.

Dataset

- Tables used: shipments, shipment_items, customers (optional: carriers, warehouses, tracking_events).

Method (SQL Workflow)

- Filter to DELIVERED shipments to ensure completed outcomes.
- Create v_shipment_profit (shipment-level): revenue proxy = $\text{sum}(\text{qty} \times \text{unit_price})$, weight = $\text{sum}(\text{qty} \times \text{unit_weight_kg})$, ship_cost = base_cost + fuel_surcharge, gross_profit = revenue - ship_cost.
- Aggregate to Customer Scorecard: delivered_shipments, revenue, ship_cost, gross_profit, profit_margin, cost_per_kg.
- Rank Top/Bottom customers using ROW_NUMBER() window function.
- Analyze Late impact (is_late) and Monthly profit trend + cumulative profit using SUM() OVER.

Key SQL Skills Demonstrated

- JOINS
- Filtering
- Aggregations (SUM/COUNT/AVG)
- CTEs (WITH)
- Window Functions (ROW_NUMBER, SUM OVER)

Outputs to Showcase

- Top 10 & Bottom 10 customers by profit
- Customer scorecard (margin + cost/kg)
- Late impact table
- Monthly profit trend + running total

Recommendations

- Reprice loss-making customers; set minimum charge/minimum weight policies.
- Optimize carrier/route for customers with high late rate.
- Extend to exceptions (tracking_events) to quantify disruption impact.